

FILE INTEGRITY CHECKER DOCUMENTATION

1. Project Title

File Integrity Checker

2. Objective

To detect unauthorized changes in files using SHA-256 hashing and maintain data integrity.

3. Problem Statement

Important files can be modified by malware or unauthorized users without the owner's knowledge. This project helps detect such changes by comparing file hash values.

4. Research Findings

- SHA-256 generates a unique hash for every file.
- Even a small change in a file produces a different hash.
- Hashing is commonly used in cybersecurity for integrity verification.

5. Methodology

- Select a file.
- Generate its SHA-256 hash.
- Save the hash value.
- Recheck the file and generate a new hash.
- Compare both hashes.
- If hashes differ, the file has been modified.

6. Tools Used

- Python
- VS Code
- hashlib Library
- GitHub

7. Implementation

```
import hashlib
```

```
def calculate_hash(file_path):  
    sha256 = hashlib.sha256()
```

```
    with open(file_path, "rb") as file:  
        while chunk := file.read(4096):  
            sha256.update(chunk)
```

```
    return sha256.hexdigest()
```

```
file_path = input("Enter File Path: ")  
print(calculate_hash(file_path))
```

8. Results

- Generated SHA-256 hash values successfully.
- Detected file modifications accurately.
- Improved file integrity monitoring.

9. Future Scope

- Real-time monitoring
- Email alerts
- Multiple file verification

10. Conclusion

The File Integrity Checker is a simple cybersecurity project that helps detect unauthorized file modifications and ensures data integrity using SHA-256 hashing.